

Machine Learning I: Hands-On 2 - Worked Solutions

2.1 CART: worked reference example

The original solution develops the reference dataset step by step. We reproduce that calculation first because it shows exactly how CART evaluates candidate splits.

A_1	A_2	A_3	Y
0	0	1	1
1	0	1	1
1	1	1	1
0	0	0	0
0	1	0	1

For binary classification,

$$G = 1 - (p_0^2 + p_1^2),$$

and for a split,

$$G_{\text{split}} = \frac{n_L}{n} G_L + \frac{n_R}{n} G_R.$$

CART selects the split with the smallest weighted impurity.

Step 1: Compute impurity at the root

There are five observations. Class $Y = 1$ appears four times and class $Y = 0$ appears once, so

$$p_1 = \frac{4}{5} = 0.8, \quad p_0 = \frac{1}{5} = 0.2.$$

Hence

$$G_{\text{root}} = 1 - (0.8^2 + 0.2^2) = 1 - (0.64 + 0.04) = 0.32.$$

Step 2: Evaluate the split on A_1

For $A_1 = 0$, the observations are rows 1, 4, and 5 with targets 1, 0, 1. Thus

$$p_1 = \frac{2}{3}, \quad p_0 = \frac{1}{3},$$

and

$$G_L = 1 - \left(\frac{2}{3}\right)^2 - \left(\frac{1}{3}\right)^2 = \frac{4}{9} \approx 0.4444.$$

For $A_1 = 1$, rows 2 and 3 both have $Y = 1$, so the right child is pure:

$$G_R = 0.$$

The weighted impurity is therefore

$$G_{\text{split}}(A_1) = \frac{3}{5} \left(\frac{4}{9} \right) + \frac{2}{5}(0) = \frac{4}{15} \approx 0.2667.$$

Step 3: Evaluate the split on A_2

For $A_2 = 0$, rows 1, 2, and 4 have targets 1, 1, 0, so

$$G_L = \frac{4}{9} \approx 0.4444.$$

For $A_2 = 1$, rows 3 and 5 both have $Y = 1$, so

$$G_R = 0.$$

Thus

$$G_{\text{split}}(A_2) = \frac{3}{5} \left(\frac{4}{9} \right) + \frac{2}{5}(0) \approx 0.2667.$$

Step 4: Evaluate the split on A_3

For $A_3 = 0$, rows 4 and 5 have targets 0, 1, giving

$$G_L = 1 - \left(\frac{1}{2} \right)^2 - \left(\frac{1}{2} \right)^2 = 0.5.$$

For $A_3 = 1$, rows 1, 2, and 3 all have $Y = 1$, so

$$G_R = 0.$$

Therefore

$$G_{\text{split}}(A_3) = \frac{2}{5}(0.5) + \frac{3}{5}(0) = 0.2.$$

Step 5: Choose the best root split

The candidate weighted impurities are

$$G(A_1) \approx 0.2667, \quad G(A_2) \approx 0.2667, \quad G(A_3) = 0.2.$$

The smallest value is obtained with A_3 , so CART chooses A_3 **as the root split**.

The branch $A_3 = 1$ is already pure and predicts $Y = 1$. We only need to continue with the mixed branch $A_3 = 0$.

Step 6: Continue splitting the node $A_3 = 0$

This subset contains rows 4 and 5:

Row	A_1	A_2	A_3	Y
4	0	0	0	0
5	0	1	0	1

At this node, A_1 and A_3 are constant, so neither can separate the two observations. Only A_2 changes.

Splitting on A_2 gives

- $A_2 = 0$: one observation with $Y = 0$, impurity 0;
- $A_2 = 1$: one observation with $Y = 1$, impurity 0.

The weighted impurity is therefore

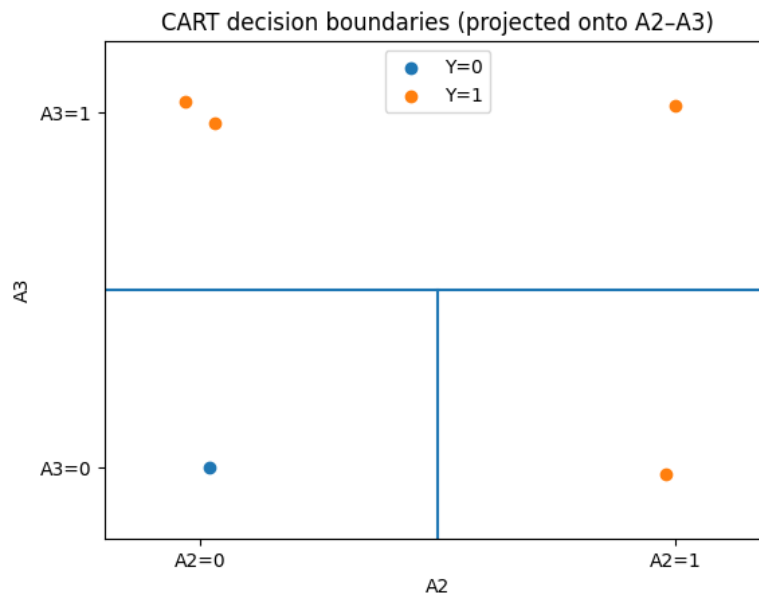
$$\frac{1}{2}(0) + \frac{1}{2}(0) = 0.$$

This is a perfect separation.

Final learned tree

The resulting rules are:

1. If $A_3 = 1$, predict $Y = 1$.
2. If $A_3 = 0$ and $A_2 = 0$, predict $Y = 0$.
3. If $A_3 = 0$ and $A_2 = 1$, predict $Y = 1$.



ML insight: greedy impurity reduction

CART chooses the best split **at the current node**. It does not directly optimize the final tree as one global object. In this example, A_3 produces the largest immediate reduction in impurity. Once that split is made, only the remaining mixed region needs another partition.

2.2 Applying the same procedure to the exercise dataset

For the exercise dataset, the target values are 0, 0, 0, 1, 1, so

$$G_{\text{root}} = 1 - \left(\frac{3}{5}\right)^2 - \left(\frac{2}{5}\right)^2 = 0.48.$$

The weighted root impurities are

$$G(A_1) = 0.40, \quad G(A_2) \approx 0.2667, \quad G(A_3) \approx 0.4667.$$

Therefore the best root split is A_2 .

The branch $A_2 = 0$ is pure and predicts class 0. In the branch $A_2 = 1$, the three remaining observations are

Example	A_1	A_3	Y
3	0	0	0
4	1	1	1
5	1	0	1

Splitting this node on A_1 makes both children pure, giving weighted impurity 0. The resulting rules are therefore

- if $A_2 = 0$, predict $Y = 0$;
- if $A_2 = 1$ and $A_1 = 0$, predict $Y = 0$;
- if $A_2 = 1$ and $A_1 = 1$, predict $Y = 1$.

